

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT1 — EXPERIMENT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	Assessment:	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data. (BL3)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	11.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	15%	The chart is neatly and	The chart is well organized, with	Basic construction;	Poorly constructed; the

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	only minor issues in labeling, legend, or layout.	required elements (title, labels, legend) are present but poorly arranged or partly missing.	chart is missing key elements such as a title, labels, or a legend.
Accuracy	10%	The chart is completely accurate — all values, proportions, quartiles, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5

Data: A company's revenue by product line: Electronics = ₹50,000, Clothing = ₹30,000, Books = ₹20,000.

Task: Construct a pie chart showing each product line's percentage share of total revenue.

Calculation

The percentage share is:

Percentage=product Revenue/total Revenue*100

Therefore:

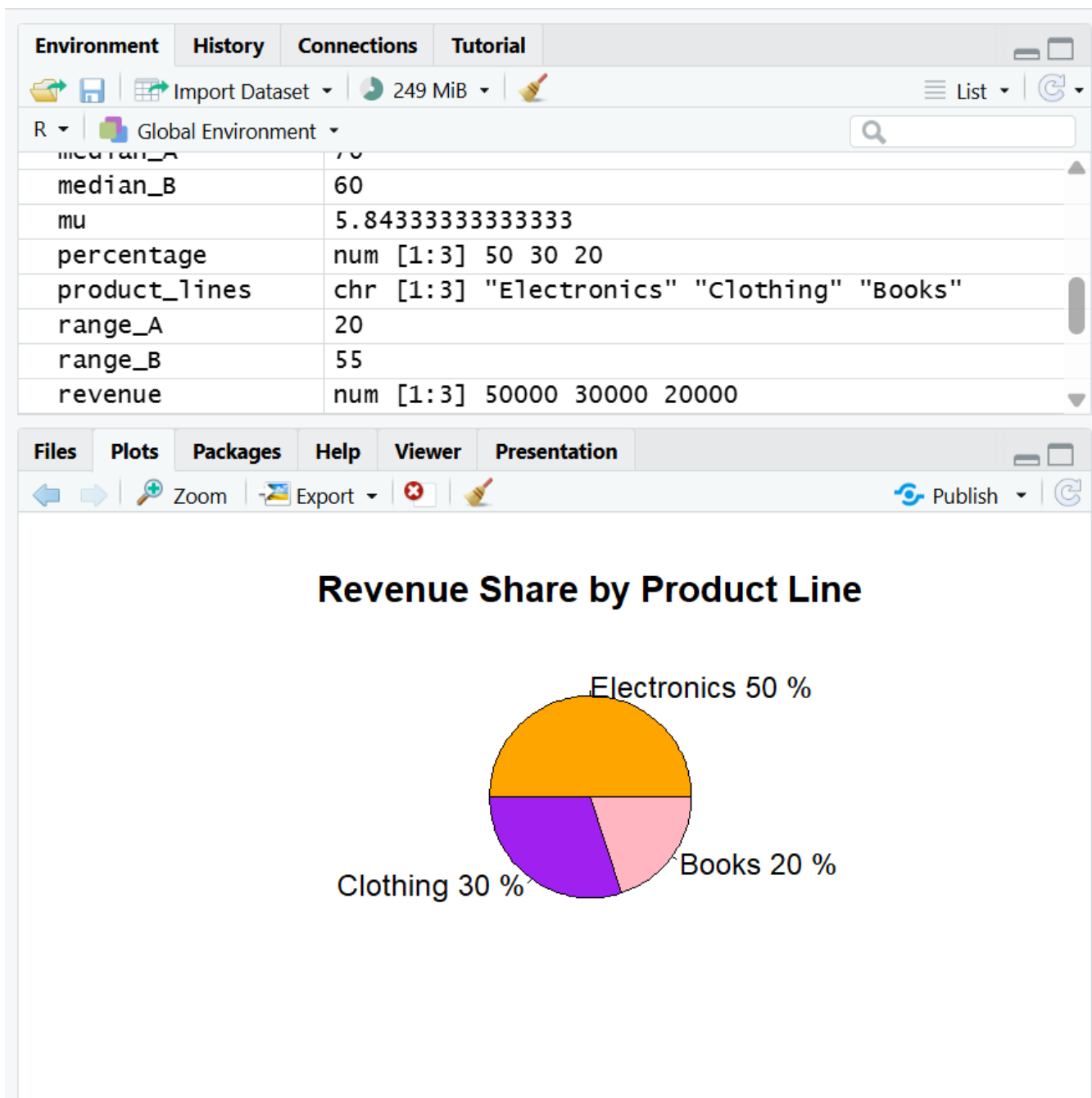
- Electronics = $50000/100000*100=50\%$
- Clothing = $30000/100000*100=30\%$
- Books = $20000/100000*100=20\%$

```
# Data: Company's revenue by product line
product_lines <- c("Electronics", "Clothing", "Books")
revenue <- c(50000, 30000, 20000)

# Calculate percentage share
percentage <- revenue / sum(revenue) * 100

# Display percentages
print(percentage)

# Pie chart
pie(revenue,
    labels = paste(product_lines,
                    round(percentage, 1), "%"),
    main = "Revenue Share by Product Line",
    col = c("orange", "purple", "lightpink"))
```

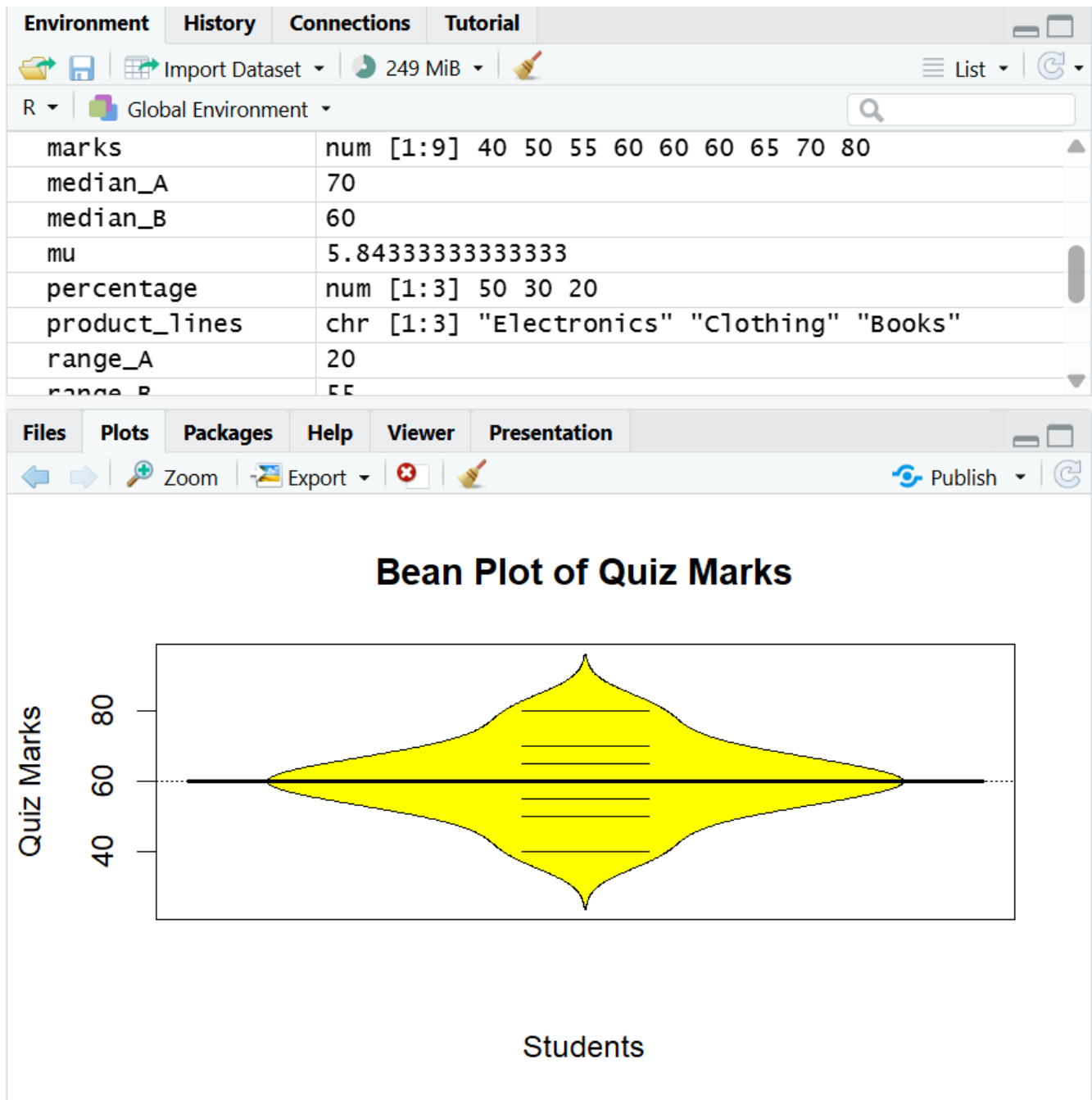


Problem 6

Data: Quiz marks of 9 students: 40, 50, 55, 60, 60, 60, 65, 70, 80.

Task: Construct a bean plot for this data and identify any repeated values that become visible in the plot.

```
1 marks <- c(40, 50, 55, 60, 60, 60, 65, 70, 80)
2 # Install beanplot package if not already installed
3 # install.packages("beanplot")
4 library(beanplot)
5 # Construct bean plot
6 beanplot(marks,
7           main = "Bean Plot of Quiz Marks",
8           ylab = "Quiz Marks",
9           xlab = "Students",
10          col = "yellow")
11 # Identify repeated values
12 print("Repeated values:")
13 print(unique(marks[duplicated(marks)]))
14 # Frequency of each mark
15 print("Frequency of marks:")
16 print(table(marks))
```



Problem 12

Data: Two classes' marks — Class A (clustered): 50, 55, 58, 60, 60, 62, 65.

Class B (spread): 30, 40, 50, 60,

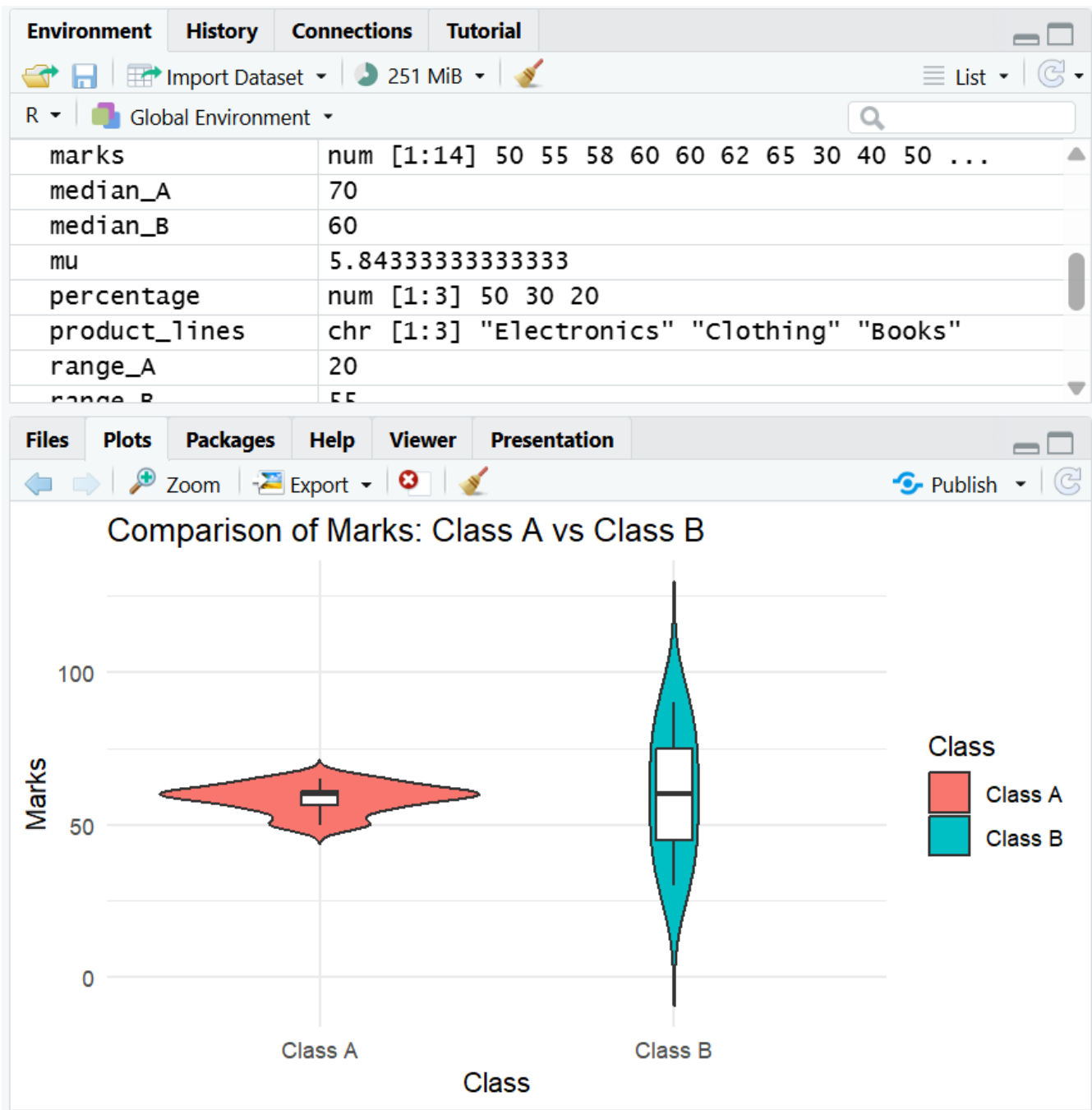
70, 80, 90.

Task: Construct violin plots for both classes and compare their shapes, even though both classes have a similar average mark.

```

1 # Class A marks - clustered
2 class_A <- c(50, 55, 58, 60, 60, 62, 65)
3 # Class B marks - spread
4 class_B <- c(30, 40, 50, 60, 70, 80, 90)
5 # Combine data
6 marks <- c(class_A, class_B)
7 class_name <- c(rep("Class A", length(class_A)),
8                 rep("Class B", length(class_B)))
9 # Create data frame
10 data <- data.frame(
11   Class = class_name,
12   Marks = marks
13 )
14 # Install ggplot2 if not already installed
15 # install.packages("ggplot2")
16 library(ggplot2)
17 # Construct violin plots
18 geom_violin(trim = FALSE) +
19 geom_boxplot(width = 0.1, fill = "white") +
20 labs(
21   title = "Comparison of Marks: Class A vs Class B",
22   x = "Class",
23   y = "Marks"
24 ) +
25 theme_minimal()
26 # Calculate and display averages
27 print("Mean of Class A:")
28 print(mean(class_A))
29 print("Mean of Class B:")
30 print(mean(class_B))
31

```

Expected Mean

For Class A: $50+55+58+60+60+62+65/7=58.57$

For Class B: $30+40+50+60+70+80+90/7=60$

So the means are approximately:

Mean of Class A = 58.57

Mean of Class B = 60

Problem 18

Data: Test scores for two sections — Section A: 60, 65, 68, 70, 72, 75, 80.

Section B: 40, 50, 55, 60, 65,
70, 95.

Task: Construct box plots for both sections and compare their medians and spread.

```
1 # Section A scores
2 section_A <- c(60, 65, 68, 70, 72, 75, 80)
3 # Section B scores
4 section_B <- c(40, 50, 55, 60, 65, 70, 95)
5 # Combine data
6 scores <- c(section_A, section_B)
7 section <- c(rep("Section A", length(section_A)),
8             rep("Section B", length(section_B)))
9 # Create data frame
10 data <- data.frame(
11   Section = section,
12   Score = scores
13 )
14 # Construct box plots
15 boxplot(Score ~ Section,
16         data = data,
17         main = "Box Plot of Test Scores",
```

```
19   ylab = "Test Scores",
20   col = c("grey", "violet"))
21
22 # Calculate medians
23 median_A <- median(section_A)
24 median_B <- median(section_B)
25 # Calculate spreads
26 range_A <- max(section_A) - min(section_A)
27 range_B <- max(section_B) - min(section_B)
28 # Display results
29 print("Median of Section A:")
30 print(median_A)
31 print("Median of Section B:")
32 print(median_B)
33 print("Range of Section A:")
34 print(range_A)
35
36 print("Range of Section B:")
```

37:15 (Top Level) ↕

R Script

Environment

History

Connections

Tutorial

Import Dataset

251 MiB

List

R

Global Environment

range_b	55
revenue	num [1:3] 50000 30000 20000
scores	num [1:14] 60 65 68 70 72 75 80 40 50 55 ...
section	chr [1:14] "Section A" "Section A" "Section A"...
section_A	num [1:7] 60 65 68 70 72 75 80
section_B	num [1:7] 40 50 55 60 65 70 95
sigma	0.828066127977863
x	num [1:401] 4 4.01 4.02 4.03 4.04 4.05 4.06 4...

